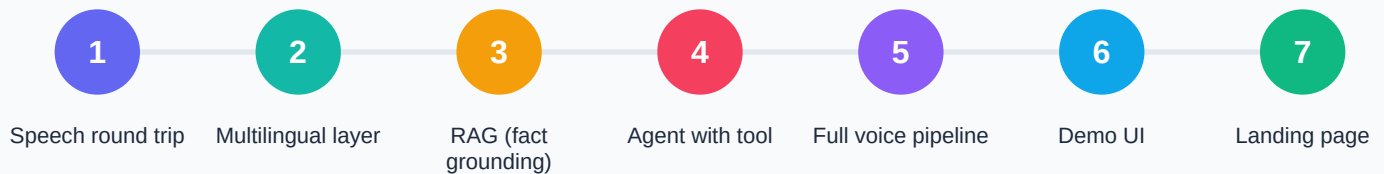


Multilingual Voice Assistant

Project roadmap, phase by phase

Indian fintech rules & schemes assistant

This project is a voice assistant that speaks Hindi, Kannada, Tamil, and English.
It was built step by step, in seven small phases.
Each page in this document explains one phase in plain, simple language.



How to read each phase page

WHY

Why this phase is needed

WHAT

What actually gets built

OUTCOME

What you get at the end

Speech round trip

Getting voice in and out working

WHY THIS PHASE

First, we must check the basics. Voice input and voice output are the core of this whole project. If this simple loop fails, nothing else will work.

WHAT WE DO

We record a short voice clip from the microphone. Whisper turns that speech into text. The same text is then spoken back out loud.

OUTCOME

We get a small working program. It listens, understands English speech, and replies out loud. This proves the base pipeline is solid.

Multilingual layer

Understanding Indian languages

WHY THIS PHASE

Real users do not always speak English. Many people feel more comfortable in Hindi, Kannada, or Tamil. The assistant needs to meet them in their own language.

WHAT WE DO

We detect which language the person is speaking. Their words are translated into English to process. The final answer is translated back and spoken aloud.

OUTCOME

The assistant can now understand and reply in four languages. This makes it far more useful for real Indian users.

RAG (fact grounding)

Making answers true, not guessed

WHY THIS PHASE

A voice assistant that makes up facts is risky, especially for money and rules. Answers about regulation must be accurate. Guessing is not good enough here.

WHAT WE DO

We store real FAQ information in a searchable index. For each question, we fetch the most relevant facts first. The answer is written using only those facts.

OUTCOME

The assistant now gives grounded, trustworthy answers. Every reply can be traced back to a real source document.

Agent with tool

Letting the assistant take action

WHY THIS PHASE

Some questions need more than plain information. A user may want to check something specific about themselves. The assistant should be able to act, not just talk.

WHAT WE DO

We build a small decision-maker using LangGraph. It reads the question and decides where to send it. One path answers from FAQs, the other runs an eligibility check.

OUTCOME

The assistant becomes genuinely agentic. It picks the right tool for the job, on its own, every time.

Full voice pipeline

Connecting every piece

WHY THIS PHASE

Until now, each piece was tested alone. A real user does not care about phases. They just want one smooth experience, start to end.

WHAT WE DO

We connect speech recognition, translation, the agent, and text-to-speech into one flow. A single spoken question goes in. A single spoken answer comes out.

OUTCOME

The full assistant now works end to end. Anyone can speak a question and hear a real, grounded answer.

WHY THIS PHASE

A terminal script is great for testing but hard to show off. Other people need something simple and visual. A demo should feel effortless to try.

WHAT WE DO

We wrap the full pipeline in a small Gradio web app. It has one microphone button and plays the spoken answer back. No coding knowledge is needed to use it.

OUTCOME

There is now a clean, browser-based demo. It is ready to share on a call or during an interview.

Landing page

Sharing the project online

WHY THIS PHASE

A project also needs a public face. Code alone does not explain itself to a recruiter scrolling LinkedIn. A short, clear page tells the story fast.

WHAT WE DO

We build a small React landing page. It shows the project name, the five-step pipeline, and the key features. It is styled to look clean and modern.

OUTCOME

There is now a shareable page for the project. It works well as a screenshot or as a live deployed link.